

PRD: Market Instinct — Real-Time Prediction Game

Owner: Sandeep | Priority: P0 (new product) | Target: MVP in 4-6 weeks

1. What and Why

A standalone, free-to-play prediction game where users call **Long / Short / No Trade** on real, world-market charts (crypto, forex, indices, stocks) and get scored on accuracy — not just entertainment, but a lightweight instinct-calibration tool for retail traders. Free tier runs on delayed data; premium unlocks true real-time streaming and full asset coverage. This is a separate product from TradeSaath, sharing no codebase or brand.

Why now: proven category (Tradiko, ChartGame, Tradicted all validate demand), cheap to build an MVP of, and it's a low-friction acquisition funnel — a 2-minute game is a much easier first touch than a full psychology platform.

2. Target User & Differentiation

Persona: retail traders and aspiring traders worldwide, not India-specific. They already know Tradiko-style games; the reason to switch/stay is that this one **teaches**, not just entertains.

Differentiation from Tradiko/ChartGame/Tradicted (the reason this isn't a dead clone):

- Post-round breakdown: what actually happened after the call (follow-through candles, volume context) — not just win/loss
- Accuracy tracked **per instrument class and per timeframe**, so a user learns "I read forex well, I guess randomly on crypto" — a real signal
- Difficulty/round selection by **volatility regime**, not one flat leaderboard
- True world-market coverage (forex + indices + stocks + crypto), not crypto-only

3. MVP Scope

IN (V1):

- Core loop: show chart segment → user picks Long/Short/No Trade → reveal outcome → score
- Asset classes: crypto (free, real-time via exchange WS) + forex majors + 2-3 global indices (delayed, free tier)
- Free tier: 15-min delayed data, limited asset list, capped rounds/day
- Premium tier: real-time data, full asset list, unlimited rounds
- Per-instrument, per-timeframe accuracy tracking
- Simple global leaderboard (daily + all-time)

- Auth (needed to persist accuracy history across sessions)
- Payment integration for premium (Razorpay, matching your existing stack knowledge)

OUT (V2+):

- Stocks beyond a small curated list (full 125K+ symbol coverage — cost-gated)
- Social features (friend challenges, sharing)
- Mobile app (web-first, responsive)
- AI-generated post-round commentary
- Skill tree / XP / streak gamification layers
- Admin panel (use Supabase dashboard directly for V1)

The one thing: if a user can only do one thing, it's play a round and see *why* their call was right or wrong. Everything else is scaffolding around that.

4. Architecture

Data layer (hybrid — see prior discussion)

- **Crypto:** direct exchange WebSocket feeds (Binance public WS) — free, truly real-time, no vendor cost
- **Forex/indices/stocks:** FCS API (~\$10/mo tier) for MVP — bundles forex, crypto backup, and stock symbols in one plan; free tier (500 calls/mo) usable for initial dev/testing
- **Free vs premium gate:** free tier serves 15-min-delayed snapshots (pulled and cached server-side on an interval); premium tier gets the live WS stream directly

Services

- **Next.js app** (Vercel): UI, auth, game logic, leaderboard, payments — same pattern you already run for TradeSaath
- **Standalone price-streaming service** (Railway/Fly.io, small always-on Node process): holds persistent WebSocket connections to Binance + FCS API, normalizes ticks, fans out to connected clients over its own WS endpoint. This is the one piece that does NOT fit Vercel serverless — flagging it now so it's not a surprise mid-build.
- **Supabase:** users, round history, accuracy stats, leaderboard, RLS per user

Data Changes

Table: rounds

```

id UUID PK DEFAULT gen_random_uuid()
user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE
instrument TEXT NOT NULL          -- e.g. 'BTC/USD', 'EUR/USD'
instrument_class TEXT NOT NULL CHECK (instrument_class IN
('crypto','forex','index','stock'))
timeframe TEXT NOT NULL          -- e.g. '1m','5m','15m'
call TEXT NOT NULL CHECK (call IN ('long','short','no_trade'))
outcome TEXT CHECK (outcome IN ('correct','incorrect','neutral'))
entry_price NUMERIC NOT NULL
resolve_price NUMERIC
created_at TIMESTAMPTZ DEFAULT NOW()
resolved_at TIMESTAMPTZ

```

Index: idx_rounds_user_instrument ON rounds(user_id, instrument_class, timeframe)

RLS: Users can only read/write their own rounds

Table: leaderboard_daily (materialized/aggregated, refreshed on a cron)

```

user_id UUID
date DATE
correct_count INT
total_count INT
accuracy NUMERIC GENERATED ALWAYS AS (correct_count::numeric /
NULLIF(total_count,0)) STORED

```

API (Next.js app)

POST /api/rounds/start

Auth: required

Input: { instrumentClass: string, timeframe: string }

Output 200: { roundId, instrument, chartData, entryPrice }

Output 403: { error: "Free tier asset limit reached" }

POST /api/rounds/:id/call

Auth: required

Input: { call: 'long' | 'short' | 'no_trade' }

Output 200: { outcome, resolvePrice, followThroughData }

GET /api/stats/accuracy

Auth: required

Output 200: { byInstrumentClass: {...}, byTimeframe: {...}, overall: number }

```
GET /api/leaderboard?period=daily|alltime
Auth: public
Output 200: { entries: [{ userId, displayName, accuracy, roundsPlayed }] }
```

Price-streaming service (separate WS endpoint)

```
WS wss://stream.<domain>/prices
Auth: token (free = throttled/delayed channel, premium = live channel)
Server → Client: { instrument, price, timestamp }
```

5. UI States

- Landing: instrument class picker (crypto/forex/index), "Play free round" CTA
- Round in progress: chart with countdown, Long/Short/No Trade buttons
- Reveal: outcome animation, entry vs resolve price, follow-through mini-chart
- Empty stats: "Play your first round to start building your accuracy profile"
- Upgrade prompt: shown when free asset/round cap is hit, not blocking mid-round
- Error (data feed down): "Market data temporarily unavailable" + retry, never a silent frozen chart

6. Monetization

- Free: delayed data, capped daily rounds, crypto + majors only
- Premium (subscription, Razorpay): real-time data, unlimited rounds, full asset list, extended accuracy history
- No ads — keeps the product credible to traders who'd view ads as noise

7. Phased Roadmap

- **Phase 1 (MVP, 4-6 wks):** core loop, crypto + forex majors, free tier only, basic leaderboard, auth — validate the loop is fun/sticky before adding payments
- **Phase 2:** premium tier + real-time gate + payments, expand instrument list
- **Phase 3:** per-instrument/timeframe analytics dashboard, volatility-regime difficulty selection
- **Phase 4 (maybe):** social/challenge features, mobile wrapper

8. Open Questions

- Domain/brand name — not yet decided

- Exact free-tier round cap (needs a number before build: suggest starting at 20 rounds/day and tuning from usage data)
- Whether "No Trade" calls should count toward accuracy or be excluded entirely (excluding avoids gaming the leaderboard by spamming No Trade)
- Legal: confirm 15-min-delay free tier requires no special licensing per data vendor's terms before launch